

Chapter 11

Circle

Only One Option Correct Type Questions

1. Consider the two curves

$$C_1 : y^2 = 4x$$

$$C_2 : x^2 + y^2 - 6x + 1 = 0$$

then,

[IIT-JEE-2008 (Paper-1)]

- (A) C_1 and C_2 touch each other only at one point
(B) C_1 and C_2 touch each other exactly at two points
(C) C_1 and C_2 intersect (but do not touch) at exactly two points
(D) C_1 and C_2 neither intersect nor touch each other

2. Tangents drawn from the point $P(1, 8)$ to the circle

$$x^2 + y^2 - 6x - 4y - 11 = 0$$

touch the circle at the point A and B . The equation of the circumcircle of the triangle PAB is

[IIT-JEE-2009 (Paper-1)]

- (A) $x^2 + y^2 + 4x - 6y + 19 = 0$ (B) $x^2 + y^2 - 4x - 10y + 19 = 0$
(C) $x^2 + y^2 - 2x + 6y - 29 = 0$ (D) $x^2 + y^2 - 6x - 4y + 19 = 0$

3. The circle passing through the point $(-1, 0)$ and touching the y -axis at $(0, 2)$ also passes through the point

[IIT-JEE-2011 (Paper-2)]

- (A) $\left(-\frac{3}{2}, 0\right)$ (B) $\left(-\frac{5}{2}, 2\right)$
(C) $\left(-\frac{3}{2}, \frac{5}{2}\right)$ (D) $(-4, 0)$

4. The locus of the mid-point of the chord of contact of tangents drawn from points lying on the straight line $4x - 5y = 20$ to the circle $x^2 + y^2 = 9$ is

[IIT-JEE-2012 (Paper-1)]

- (A) $20(x^2 + y^2) - 36x + 45y = 0$ (B) $20(x^2 + y^2) + 36x - 45y = 0$
(C) $36(x^2 + y^2) - 20x + 45y = 0$ (D) $36(x^2 + y^2) + 20x - 45y = 0$

5. A line $y = mx + 1$ intersects the circle $(x - 3)^2 + (y + 2)^2 = 25$ at the points P and Q . If the midpoint of the line segment PQ has x -coordinate $-\frac{3}{5}$, then which one of the following options is correct?

[JEE (Adv)-2019 (Paper-1)]

- (A) $2 \leq m < 4$ (B) $4 \leq m < 6$
(C) $6 \leq m < 8$ (D) $-3 \leq m < -1$

One or More Option(s) Correct Type Questions

6. Circle(s) touching x -axis at a distance 3 from the origin and having an intercept of length $2\sqrt{7}$ on y -axis is (are) [JEE (Adv)-2013 (Paper-2)]
- (A) $x^2 + y^2 - 6x + 8y + 9 = 0$ (B) $x^2 + y^2 - 6x + 7y + 9 = 0$
 (C) $x^2 + y^2 - 6x - 8y + 9 = 0$ (D) $x^2 + y^2 - 6x - 7y + 9 = 0$
7. A circle S passes through the point $(0, 1)$ and is orthogonal to the circles $(x - 1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$. Then [JEE (Adv)-2014 (Paper-1)]
- (A) Radius of S is 8 (B) Radius of S is 7
 (C) Centre of S is $(-7, 1)$ (D) Centre of S is $(-8, 1)$
8. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P, Q are functions of x, y and $y' \left(\text{here } y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2} \right)$, then which of the following statements is(are) true? [JEE (Adv)-2015 (Paper-1)]
- (A) $P = y + x$ (B) $P = y - x$
 (C) $P + Q = 1 - x + y + y' + (y')^2$ (D) $P - Q = x + y - y' - (y')^2$
9. The circle $C_1 : x^2 + y^2 = 3$, with centre at O , intersects the parabola $x^2 = 2y$ at the point P in the first quadrant. Let the tangent to the circle C_1 at P touches other two circles C_2 and C_3 at R_2 and R_3 , respectively. Suppose C_2 and C_3 have equal radii $2\sqrt{3}$ and centers Q_2 and Q_3 , respectively. If Q_2 and Q_3 lie on the y -axis, then [JEE (Adv)-2016 (Paper-1)]
- (A) $Q_2Q_3 = 12$ (B) $R_2R_3 = 4\sqrt{6}$
 (C) Area of the triangle OR_2R_3 is $6\sqrt{2}$ (D) Area of the triangle PQ_2Q_3 is $4\sqrt{2}$
10. Let RS be the diameter of the circle $x^2 + y^2 = 1$, where S is the point $(1, 0)$. Let P be a variable point (other than R and S) on the circle and tangents to the circle at S and P meet at the point Q . The normal to the circle at P intersects a line drawn through Q parallel to RS at point E . Then the locus of E passes through the point(s). [JEE (Adv)-2016 (Paper-1)]
- (A) $\left(\frac{1}{3}, \frac{1}{\sqrt{3}} \right)$ (B) $\left(\frac{1}{4}, \frac{1}{2} \right)$
 (C) $\left(\frac{1}{3}, -\frac{1}{\sqrt{3}} \right)$ (D) $\left(\frac{1}{4}, -\frac{1}{2} \right)$
11. Let T be the line passing through the points $P(-2, 7)$ and $Q(2, -5)$. Let F_1 be the set of all pairs of circles (S_1, S_2) such that T is tangent to S_1 at P and tangent to S_2 at Q , and also such that S_1 and S_2 touch each other at a point, say, M . Let E_1 be the set representing the locus of M as the pair (S_1, S_2) varies in F_1 . Let the set of all straight line segments joining a pair of distinct points of E_1 and passing through the point $R(1, 1)$ be F_2 . Let E_2 be the set of the mid-points of the line segments in the set F_2 . Then, which of the following statement(s) is (are) TRUE? [JEE (Adv)-2018 (Paper-2)]
- (A) The point $(-2, 7)$ lies in E_1 (B) The point $\left(\frac{4}{5}, \frac{7}{5} \right)$ does NOT lie in E_2
 (C) The point $\left(\frac{1}{2}, 1 \right)$ lies in E_2 (D) The point $\left(0, \frac{3}{2} \right)$ does NOT lie in E_1

Linked Comprehension Type Questions

Paragraph for Q.Nos. 12 to 14

A circle C of radius 1 is inscribed in an equilateral triangle PQR . The points of contact of C with the sides PQ , QR , RP are D , E , F respectively. The line PQ is given by the equation $\sqrt{3}x + y - 6 = 0$ and the point D is $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$. Further, it is given that the origin and the centre of C are on the same side of the line PQ .

[IIT-JEE-2008 (Paper-1)]

Choose the correct answer :

12. The equation of circle C is

- (A) $(x - 2\sqrt{3})^2 + (y - 1)^2 = 1$ (B) $(x - 2\sqrt{3})^2 + \left(y + \frac{1}{2}\right)^2 = 1$
 (C) $(x - \sqrt{3})^2 + (y + 1)^2 = 1$ (D) $(x - \sqrt{3})^2 + (y - 1)^2 = 1$

13. Points E and F are given by

- (A) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), (\sqrt{3}, 0)$ (B) $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right), (\sqrt{3}, 0)$
 (C) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

14. Equations of the sides QR , RP are

- (A) $y = \frac{2}{\sqrt{3}}x + 1, y = -\frac{2}{\sqrt{3}}x - 1$ (B) $y = \frac{1}{\sqrt{3}}x, y = 0$
 (C) $y = \frac{\sqrt{3}}{2}x + 1, y = -\frac{\sqrt{3}}{2}x - 1$ (D) $y = \sqrt{3}x, y = 0$

Paragraph for Q.Nos. 15 and 16

A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3}, 1)$. A straight line L , perpendicular to PT is a tangent to the circle $(x - 3)^2 + y^2 = 1$.

[IIT-JEE-2012 (Paper-2)]

Choose the correct answer :

15. A von tangent of the two circles is

- (A) $x = 4$ (B) $y = 2$
 (C) $x + \sqrt{3}y = 4$ (D) $x + 2\sqrt{2}y = 6$

16. A possible equation of L is

- (A) $x - \sqrt{3}y = 1$ (B) $x + \sqrt{3}y = 1$
 (C) $x - \sqrt{3}y = -1$ (D) $x + \sqrt{3}y = 5$

Paragraph for Q.Nos. 17 and 18

Let S be the circle in the xy -plane defined by the equation $x^2 + y^2 = 4$.

[JEE (Adv)-2018 (Paper-1)]

Choose the correct answer :

17. Let E_1E_2 and F_1F_2 be the chords of S passing through the point $P_0(1, 1)$ and parallel to the x -axis and the y -axis, respectively. Let G_1G_2 be the chord of S passing through P_0 and having slope -1 . Let the tangents to S at E_1 and E_2 meet at E_3 , the tangents to S at F_1 and F_2 meet at F_3 , and the tangents to S at G_1 and G_2 meet at G_3 . Then, the points E_3 , F_3 , and G_3 lie on the curve
- (A) $x + y = 4$ (B) $(x - 4)^2 + (y - 4)^2 = 16$
 (C) $(x - 4)(y - 4) = 4$ (D) $xy = 4$
18. Let P be a point on the circle S with both coordinates being positive. Let the tangent to S at P intersect the coordinate axes at the points M and N . Then, the mid-point of the line segment MN must lie on the curve
- (A) $(x + y)^2 = 3xy$ (B) $x^{2/3} + y^{2/3} = 2^{4/3}$
 (C) $x^2 + y^2 = 2xy$ (D) $x^2 + y^2 = x^2y^2$

Matrix-Match / Matrix Based Type Questions

19. Answer the following by appropriately matching the lists based on the information given in the paragraph.
- Let the circles $C_1 : x^2 + y^2 = 9$ and $C_2 : (x - 3)^2 + (y - 4)^2 = 16$, intersect at the points X and Y . Suppose that another circle $C_3 : (x - h)^2 + (y - k)^2 = r^2$ satisfies the following conditions :
- (i) centre of C_3 is collinear with the centres of C_1 and C_2 ,
 (ii) C_1 and C_2 both lie inside C_3 , and
 (iii) C_3 touches C_1 at M and C_2 at N .
- Let the line through X and Y intersect C_3 at Z and W , and let a common tangent of C_1 and C_3 be a tangent to the parabola $x^2 = 8\alpha y$.

There are some expressions given in the List-I whose values are given in List-II below :

[JEE(Advanced)-2019 (Paper-2)]

List-I	List-II
(I) $2h + k$	(P) 6
(II) $\frac{\text{Length of } ZW}{\text{Length of } XY}$	(Q) $\sqrt{6}$
(III) $\frac{\text{Area of triangle } MZN}{\text{Area of triangle } ZMW}$	(R) $\frac{5}{4}$
(IV) α	(S) $\frac{21}{5}$
	(T) $2\sqrt{6}$
	(U) $\frac{10}{3}$

Which of the following is the only CORRECT combination?

- (A) (I), (S) (B) (II), (Q)
 (C) (II), (T) (D) (I), (U)

20. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Let the circles $C_1 : x^2 + y^2 = 9$ and $C_2 : (x - 3)^2 + (y - 4)^2 = 16$, intersect at the points X and Y . Suppose that another circle $C_3 : (x - h)^2 + (y - k)^2 = r^2$ satisfies the following conditions:

- (i) centre of C_3 is collinear with the centres of C_1 and C_2 ,
- (ii) C_1 and C_2 both lie inside C_3 , and
- (iii) C_3 touches C_1 at M and C_2 at N .

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	(U) $\frac{10}{3}$

Which of the following is the only INCORRECT combination?

- (A) (III), (R)
- (B) (IV), (U)
- (C) (IV), (S)
- (D) (I), (P)

Integer / Numerical Value Type Questions

21. The centres of two circles C_1 and C_2 each of unit radius are at a distance of 6 units from each other. Let P be the mid point of the line segment joining the centres of C_1 and C_2 and C be a circle touching circles C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangent to C_2 and C , then the radius of the circle C is

[IIT-JEE-2009 (Paper-2)]

22. Two parallel chords of a circle of radius 2 are at a distance $\sqrt{3} + 1$ apart. If the chords subtend at the center,

angles of $\frac{\pi}{k}$ and $\frac{2\pi}{k}$, where $k > 0$, then the value of $[k]$ is

[IIT-JEE-2010 (Paper-2)]

[Note : $[k]$ denotes the largest integer less than or equal to k]

23. The straight line $2x - 3y = 1$ divides the circular region $x^2 + y^2 \leq 6$ into two parts. If

$S = \left\{ \left(2, \frac{3}{4} \right), \left(\frac{5}{2}, \frac{3}{4} \right), \left(\frac{1}{4}, -\frac{1}{4} \right), \left(\frac{1}{8}, \frac{1}{4} \right) \right\}$, then the number of point(s) in S lying inside the smaller part is

[IIT-JEE-2011 (Paper-2)]

24. For how many values of p , the circle $x^2 + y^2 + 2x + 4y - p = 0$ and the coordinate axes have exactly three common points?
[JEE (Adv)-2017 (Paper-1)]
25. Let the point B be the reflection of the point $A(2, 3)$ with respect to the line $8x - 6y - 23 = 0$. Let Γ_A and Γ_B be circles of radii 2 and 1 with centres A and B respectively. Let T be a common tangent to the circles Γ_A and Γ_B such that both the circles are on the same side of T . If C is the point of intersection of T and the line passing through A and B , then the length of the line segment AC is _____.
[JEE (Adv)-2019 (Paper-1)]
26. Let O be the centre of the circle $x^2 + y^2 = r^2$, where $r > \frac{\sqrt{5}}{2}$. Suppose PQ is a chord of this circle and the equation of the line passing through P and Q is $2x + 4y = 5$. If the centre of the circumcircle of the triangle OPQ lies on the line $x + 2y = 4$, then the value of r is _____.
[JEE (Adv)-2020 (Paper-2)]

